Associate Editor

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Comments to Author:
This paper examines the link between the gender mix of clinical teams and outcomes for obstetric patients. The authors suggest that this context is particularly well suited for studying the potential causal effect of gender dynamics on team performance. I have shared the paper with two expert reviewers and have done a detailed read myself. Both reviewers see many strengths in the work including a compelling set of research questions about gender mix and team performance, unique data, and thorough empirical analysis. Alas, both reviewers also express concerns with aspects of the paper's design and execution. My own assessment of the paper matches up closely with the strengths and concerns identified in these excellent reviews. Given the high quality of the reviews, I have relatively little additional guidance to provide the authors beyond responding to the specific points raised in the reports. That said, I highlight below several key items that emerged from their reviews and my own read of the paper.

1. Need for more detail about how team members actually interact: As the authors note on p. 7 of the manuscript (p. 9 of the full PDF), 77% of the deliveries in their sample have only one OB-GYN involved (i.e., are solo procedures). Thus, the occurrence of a team-based delivery is relatively rare. The rarity of team-based deliveries raises the question of why we would ever see the formation of OB-GYN teams in practice. The authors provide a helpful discussion of this topic on pp. 8-9 of the manuscript, where they offer two categories of explanations for team formation, which R1 refer to as: (1) clinical complexity/emergencies or (2) operational constraints (e.g., handoffs due to competing procedures or call coverage for a patient's primary OB). As R1 suggests, the inability of the authors' data to tease apart these two explanations for team formation limits their ability to distinguish between stories of team performance that assume the two team members are synchronously interacting (akin to a sports team) versus a sequential interaction between one team member who hands off to another (akin to a shift change). In my view, there could even be a third possibility that may or may not exist in the data: that two OBs are recorded in the data for a given procedure—the lead being the OB on call who performs the procedure and the assistant being the patient's primary OB—but that the work is entirely completed by the lead OB on call with no consultation with the patient's primary OB. In essence, this would be a solo delivery that, due either to recordkeeping conventions or recordkeeping errors, gets characterized as a team delivery. In short, it is not clear if what the authors are observing in their "teams" are traditional synchronous teams, handoffs, or recordkeeping anomalies. This distinction, however, is important for the reader's understanding of the potential mechanisms driving the observed outcomes.

Strategy: Mostly qualitative response using details from the data. For the team types, can point to the patterns in the low-risk vs high-risk samples.

– Not recordkeeping anomalies - to be recorded the assisting has to be directly involved in providing patient care. Also consider this for the Lead. Any summary stats on how often each doctor is solo vs team, how consistent this is across hospitals of the same type (ie this rate doesn't vary much across hospitals).

– It is more difficult to disentangle the shift change vs synchronous events with our data. Maybe the lowrisk vs highrisk cases can tell us more about this.

2. Potential for reverse causality: Particularly in the absence of information about how and when in the care delivery process an OB team is formed, the concern about reverse causality (i.e., that maternal complications might play a role in determining a team's gender mix) raised by R2 merits further consideration by the authors.

Strategy: Mostly empirical. R2 has some suggestions. This response is multi-part, starting with 1) Explaining the primary procedures & SMM - almost always a delivery. Robustness here as explained below, 2) What does the within regression capture to address this, 2) R2's suggestions around age of men results, 3) Other things…

The concern is that the second ob-gyn is only called in because of a complication & men are particularly more likely to be called in these cases. The Lead, by construction, is the first doctor assigned to the patient. We can see the procedures that assisting doctors assisted in performing in the data - XX% of these are the delivery itself. We can remove cases where the assisting may have been performing a procedure post-delivery as robustness.

Above + within physician regression should be pretty compelling

Look at SMM for outcomes that would have minimal concerns of reverse causality (but also talk about what SMM is supposed to capture) - like blood transfusion. Lean on those results.

Need to try and figure out more about when the team is "formed" in the process - qualitative + how things are coded in the data.

What other outcomes could we look at that don't have this potential concerns? I guess would just be procedures like episiotomy, forceps/vacuum etc.

3. Explanation for differences in same-gender teams: In discussing the reverse causality point, R2 observes that the paper does not clearly explain why teams with two male members consistently perform worse than the other types of teams. R1's section on "homogeneous teams" offers a similar summary point, noting: "…I found the extension of a norms-based explanation to differences between all-male and all-female teams more speculative…"

– I thought we did a pretty good job….
– Need to rethink the ordering of the discussion maybe? Based on this point and the point below.

– One proposal is to move the heterogeneity section to mechanisms and use it to explore more about what is happening with M-M. One reason is men have greater financial incentives - no. Can we say men are more aggressive in their treatment when in teams but not when solo? -

Episiotomy, vacuum/forceps, c-section (ie even if preferences don't fully explain results, they still differentially use these procedures that may be associated with greater complications)

4. Lack of significance in differences between mixed-gender (i.e., M-F vs. F-M) teams: Like both R1 and R2, I am concerned by the interpretation of the results on mixed-gender teams as suggesting "…female-led mixed-gender teams perform worse than male-led ones" (abstract). The results in Figure II(a) seem to provide no evidence of a statistically significant difference in the performance of these two types of mixed-gender teams. I don't see evidence that the authors tested for the significance of this difference and confidence intervals for these two estimates seems to overlap substantially. Like R2, I acknowledge that Figure II(c) seems to suggest that this difference is significant, yet this is presumably in a more limited sample (I could not find the results table for regression A.4 in the paper).

– Can change the framing to be more about FF> mixed gender > MM in abstract and intro. Then figure out a way to explore more of the MF vs FM later. The difference here really pops out under challenging conditions.

– This depends on the within vs between regs though, bc if reframe everything around within, then more about FF vs FM and MM vs MF.

5. Specification of main regression: I agree with R2's observation about adjusting the specification of Equation (1) on p. 10 of the manuscript. It seems that the simpler specification would include the indicator for female lead (lead_fem), an indicator for a female assisting (assist_fem), and the interaction of lead_fem and assist_fem.

This one is tricky because this is not the spec we want. Instead, I think we should propose making the four categories with MM as the leave out, which then maps easily onto the figures and gets to the core comparisons we are trying to make.

So would be some combo of within + between with the four types relative to MM -> though will need to navigate R1 streamline to only use one model concern

6. Concerns about selection in the solo-birth sample: R1 raises a compelling concern about the potential for selection in the solo-birth sample to contaminate the physician fixed effects that are used in the discussion of the mechanisms. At a minimum, this merits consideration by the authors.

Yes a central limitation. Could consider using different cuts of the solo sample based on Yiqun's suggestions.

7. Interpretation of results and strengths of claims about mechanisms: While both R1 and R2 feel that the authors have identified interesting differences in performance based on gender mix, they also raise concerns about the strength of the authors' claims regarding the gender-norm mechanisms. I will add that the authors need to be wary of using speculative language (e.g.

note the repeated use of “possibly” and the separate use “likely” in the abstract’s discussion of the findings). R1 spends a fair bit of time suggesting how the authors might improve the study in this regard.

This is very annoying. Maybe we don’t use gender norms langauge at all. Only show results the directly explain what could drive differences - ie insurance status, different procedure use, challenging conditions.

R1 doesn’t actually have suggestions, instead, it seems R1 gives us an out: “**Given the authors' claim that this is among the first papers to document how physician team gender composition affects outcomes in a medical setting, I would recommend refocusing the paper on establishing these performance differences as cleanly as possible and substantially scaling back the mechanisms claims.”**

Are there hospitals where things are better or worse and why - NFP vs FP?

How many men vs women in the hospital over time? Gender norms story here.

Differences in location of hospital and “norms”?

How to get at MM vs FF differences? Can we definitely show that men are more intensive in teams vs alone.
→ Number of procedures? What are the key procedures?
→ Episiotomy, Vaccum/Forceps, C-sections
→ Intesive preferences conditional on mode of delivery

Can we create a high-risk solo sample than looks like that team sample patients - compare behavior and preferences between these

Reviewer 1

REPORT

There is a lot to like in this paper: the setting is important and policy-relevant, the data are unusually granular and rich, and the results—especially the robust performance differences across homogeneous gender teams—are compelling. I have three main comments. First, the causal interpretation of the team effects would be strengthened by consistently adopting a more stringent treatment of hospital–time conditions governing physician availability, which—given the institutional description given in the paper—would better isolate quasi-random variation in team composition in the main analysis. Second, due to the lack of quasi-random variation in selection to patients in the solo sample, the physician fixed effects constructed from the solo sample and used in the mechanisms analysis may not be clean measures of underlying preferences or skill. Third, I think the paper may push the gender-norms explanation too strongly: differences between mixed-gender teams are relatively small and not always robust, while the norms-based interpretation of the all-female versus all-male gap is not very strongly supported. Finally, the paper would benefit from streamlining, as the large number of specifications and extensive appendix material make it difficult to identify the most credible results and the key pieces of supporting evidence.

My overall reading is that the paper can credibly establish differences in team performance across gender compositions, while the mechanisms analysis and its interpretation are weaker and, in my view, not easily fixable given the context and available data. **Given the authors' claim that this is among the first papers to document how physician team gender composition affects outcomes in a medical setting, I would recommend refocusing the paper on establishing these performance differences as cleanly as possible and substantially scaling back the mechanisms claims.** I elaborate on these points below.

1. Identification and causal claims of the first part of the analysis:

The identifying assumption is that assignment to a given team gender mix is conditionally "as-good-as-random." Based on the institutional description, two ingredients appear central for the conditional-random-assignment claim: (A) controlling for patient risk and other observable characteristics so that team types are not simply handling systematically different cases, and (B) conditioning on the relevant schedule/availability environment—i.e., the hospital–time conditions that govern which physicians are on call/backup and therefore "who happens to be available" at the moment of delivery.

Equation (1) controls extensively for patient risk - but although the paper motivates identification using rotating schedules, random labor onset, and reshuffling across shifts, the baseline specifications rely on relatively coarse hospital–time fixed effects: hospital×year and quarter-year fixed effects. While, as you note, these absorb macro changes in staffing, policies, and resources, they do not rule out selection operating at a finer within-hospital, within-year level. Team formation may still be correlated with day-of-week patterns, shift timing (night vs day), elective versus emergent delivery windows (e.g., scheduled C-sections), or periods in which only a limited subset of

physicians is on the floor. For example, if physician gender is correlated with shift coverage and off-hours deliveries are systematically riskier, hospital×year fixed effects would not address this source of selection. This tension is evident in the paper, which implicitly acknowledges it by estimating a much more stringent schedule-based specification of Equation (1) in Table 3, column 6, with hospital×(year×quarter×day-of-week×admission-hour) fixed effects. But this is only a "one-off" robustness check.

If the authors wish to maintain a causal interpretation, I think the following points should be addressed:

- Be much more explicit when describing the context/setting about the conditioning set under which assignment to team gender mix is plausibly quasi-random. In particular, clearly articulate which hospital–time availability conditions (beyond patient and physician observables) are assumed to render team formation "as-good-as-random," and how these relate to the institutional description of call schedules and shift assignment in hospitals. ✅

This and related comments below will depend on the within vs between regression narrative we adopt. Given what all the referees are saying, it seems like they still want the between but with more fixed effects as suggested, and then the within physician continues to be a secondary spec (at least for the paper - NBER presentation could be different).

But maybe a way to convince them that within is the way to start (with the additional FE) and then between - though I think we would need to show balance for both models.

- Implement the identifying specification that conditions on such set of variables consistently throughout the paper. In practice, I think this implies replicating the whole paper using the schedule-based specification in Table 3, column 6 (hospital×year×quarter×day-of-week×admission-hour fixed effects) and make it your baseline regression, as this is the specification most closely aligned with the availability-driven institutional narrative and therefore the most credible foundation for causal claims. ✅

→ Can show the gender is not related to shifts via regressions including all the interacted FE with or without other controls in both solo and team samples. This is just the nature of OB. Not just gender but can also show that teams are not specific to certain shifts.

```
reghdfe shift doc_fem_atten $doc_at if solo!=1 , absorb($fe_nd_int_nodiag) $cluster
reghdfe shift doc_fem_atten $doc_at if solo==1 , absorb($fe_nd_int_nodiag) $cluster
reghdfe shift i.team $doc_all if solo!=1, absorb($fe_nd_int_nodiag) $cluster
```

→ Can also say that using physician FE controls for these types of preferences!! Brings in the importance of the within regression. However - need to make it very clear as to why this can't be the main spec.

I did a version of this where I turned admission-hour into day shift and night shift, and day of week to weekday vs weekend and it looks good. The reason I did this is that otherwise we lose too much data (since admission time doesn't even start until 2010) and we don't have enough variation within each of these day and admission time categories – though day of week could be fine, it is mostly admission time that is the problem.. Seems this would satisfy the concerns. However, I haven't checked how this influences balance tests, and all downstream robustness checks and mechanisms.

The question then, based on this point, is what to do with the "within" physician specification vs the between specification. It seems they want this to be the between to be baseline….

- Align all the rest of the analysis with this identification strategy. Specifically: (i) Balance tests in support of quasi-random variation should be conducted using the same schedule-based conditioning set (e.g., testing whether team type or assistant gender is predictable from pre-treatment observables within hospital–time availability cells), rather than relying primarily on unconditional checks (as it is done currently – which I think is not the right way of doing it). (ii) The mechanisms analyses should, where feasible, also use the same source of variation and conditioning set as this tighter specification. ✅

  Don't 100% follow this - pretty sure our balance tests are conditional, but need to double check. Still would need to be updated using our new main spec. This point is reiterated below.

- Adopting a single, clearly articulated baseline specification that is most credible for causal inference would help not only reinforcing the causal claims but also streamline the presentation. Several current figures and appendix checks appear unnecessary (e.g., unconditional versus partially conditional balance plots), and consolidating them under one coherent identifying framework would reduce clutter and improve readability.

– This is the key comment that essentially summarizes all the prior concerns. Clearly the within is the one that meets the "most credible for causal inference" but it would change much of the focus on the paper. I am oaky with that though, in some ways, it would allow us to move away from MM vs FF and instead focus more on MF vs MM and FM vs FF. Then we could bring in MM vs FF more at the end?

2. Solo births and their role in the second part of the analysis:

A substantial portion of the mechanisms and "selection into teams tests" relies on the solo-birth sample. In particular, two central objects—physician C-section "preferences" and physician "skill"—are both constructed from solo deliveries and then used to (i) motivate the mechanisms analysis and (ii) argue against endogenous team formation by physician characteristics. This makes the interpretation of results based on the solo sample—and of the solo-based fixed effects—crucial for some of the paper's overall arguments. While I think leveraging the solo sample is potentially

very informative, the paper currently relies on it quite heavily without sufficiently discussing the associated limitations, the most important being the lack of quasi-random variation in selection to patients in the solo sample. As a result, some tests risk being over-interpreted or not very clearly explained in their nuances. The paper would benefit from being more explicit about what can be cleanly identified from solo births, what relies on stronger assumptions, and how these distinctions affect the interpretation of the mechanisms and selection results. I elaborate more below:

This is Yiqun's concern and we can try to use more exogenous solo births. See how that impacts the results.

2.1 What determines team vs solo, and what does that imply for interpretation? The institutional discussion suggests at least two distinct reasons why teams arise: (i) clinical complexity or emergencies that require additional assistance, and (ii) operational constraints such as handoffs, competing births, or the arrival of a patient's primary OB. These channels have very different implications for selection. I would therefore encourage the authors to (a) describe the team-formation process more directly and (b) document it empirically. Even a simple descriptive exercise—such as a "predict team vs solo" model using information available at admission (risk proxies, scheduled versus unscheduled indicators, time of day/day of week, hospital strain)—would help clarify whether team involvement primarily reflects clinical complexity, scheduling frictions, or a combination of the two.

This is an interesting suggestion! We could do this to learn more about what leads to solo vs team. We know for example that teams are involved in more complex patient cases.

This distinction matters because the main analysis conditions on being a team birth. If the team/solo margin is strongly endogenous to unobserved severity or operational disruptions or physician characteristics, the results are best interpreted as applying to a potentially endogenous subset of "deliveries involving two physicians" and might not extend to other type of births/context. Making this explicit seems relevant for the overall interpretation of the results.

Some practical diagnostics that could help in this direction include:

- Reporting the distribution of team involvement across physicians (the share of births delivered in teams) and whether this varies systematically by physician gender or other observable physician characteristics.
- Testing whether physician gender (and other traits) predicts the likelihood that a delivery is team versus solo, and whether this relationship varies across hospital-time conditions (e.g., time of day or day of week).
- Conditional on availability, comparing the observed prevalence of all-male, all-female, and mixed-gender teams to what would be expected under random pairing given the gender composition of on-duty physicians, as a transparent diagnostic for endogenous matching on the extensive margin.

These are all interesting suggestions. Some I think are worth doing, others seem more like they aren't understanding that we aren't looking to compare solo to team, these are fundamentally different. I guess the concern is that teams form for different reasons dependong on the gender composition of the team

2.2 Solo-sample selection and why it matters for the analysis of mechanisms: Both the "preference" and "skill" measures are physician fixed effects estimated from the solo-birth sample (with patient controls and hospital-year and quarter-year fixed effects). Due to the lack of quasi-random variation in selection to patients in the solo sample, the physician fixed effects constructed from the solo sample can easily reflect endogenous differences in solo case mix across physicians or unobserved characteristics of physician that determine such endogenous matching with patients, rather than purely intrinsic practice style ("preference") or ability ("skill"). The paper briefly acknowledges this possibility in a footnote, noting that physicians delivering solo may be able to select the panel of patients they see in ways that teams cannot. By contaminating the estimation of both FE, the endogenous nature of matching to patients in the solo-sample undermines two core claims the paper seeks to make: (i) that teams are not formed on the basis of physician skill or preferences, and (ii) that the team gender-mix effects are not simply reflecting differences in intrinsic physician practice style.

I do not think this concern has an easy or effective solution, but given how central these solo-based objects are to the mechanisms and identification arguments, I think this limitation should be discussed much more explicitly in the main text and the authors should find ways to make the analysis more transparent and credible. For example:

- Replicate the construction of the preference and/or skill measures in solo subsamples that more closely resemble the team environment (e.g., emergent cases, high-strain periods, less scheduled deliveries), and show that the measures and resulting conclusions are stable. This would directly support the implicit assumption that preferences or ability estimated in solo deliveries are informative about behavior in team deliveries despite the differences in the sample of birth in the solo vs team context.
- Be explicit about the core point that the solo-based fixed effects are potentially confounded by selection and should be interpreted as descriptive proxies rather than intrinsic physician traits. If this is the case, then the paper should correspondingly temper the part of the analysis that relies on doctors' traits estimated from the solo sample.

This all the Yiqun stuff. In addition to the emprics, we can also try to draw on other contexts that even solo deliveries are not really that endogenous. Only solo org providers would have that. Maybe we can also bring in the size of the docs primary organization and how many other docs are in that org - it would take me some time but not impossible.

2.3 Preference vs skill fixed effects: are they separable? The paper draws a conceptual and empirical distinction between (i) physician C-section "preferences" (from a solo-sample C-section regression) and (ii) physician "skill" (from a solo-sample complications regression). Both are constructed as physicians fixed effects from solo deliveries and then used for different purposes: preferences as a mechanism channel (differences in procedure choice) and skill as a selection threat (teams might be

formed by better/worse physicians). I did not understand what these two fixed effects capture in practice, and whether they are empirically and conceptually separable - especially if C-section use is an important input into maternal complications. If the authors wish to argue that the two fixed effects genuinely capture distinct concepts—preferences versus skill (setting aside the endogeneity concerns raised in Comment 2.2)—this claim would require substantially more empirical support. In particular, it would be important to demonstrate that the two objects are empirically separable and play different roles in the data.

I see, while constructfed differently - one is based on SMM and one is based on C-sections, they are likely quite linked since c-sections are highly predictive of SMM.

For example, the authors could:

- Provide diagnostics on the relationship between the two fixed effects, such as the within-physician correlation between the preference FE and the skill FE, the overlap of their distributions by gender, and whether one subsumes the other in predicting outcomes. This would help clarify whether the "preference" measure reflects a behavioral threshold distinct from ability, or whether it largely drives the "skill" proxy.
- If the evidence supports that the two fixed effects capture different dimensions, then they should be used symmetrically in the paper's key stress tests. In particular, the authors should (i) show whether the main team gender-mix results are robust to controlling for physician "skill" (not only preferences), and (ii) test whether endogenous team formation is absent not only with respect to the skill proxy but also with respect to the preference proxy.

  I think we have done this before to some extent. However, unsure how results change once we change the way the solo FE are calculated.

3. Gender norms and differences between ML-FA vs FL-MA

3.1 Mixed-gender teams: The paper places substantial emphasis on the mixed-gender lead-role contrast—female-led mixed-gender teams (FL–MA) performing worse than male-led mixed-gender teams (ML–FA)—and identifies this contrast as central to the gender-norms interpretation (i.e., that FL–MA "runs counter to traditional gender norms around leadership"). However, I am under the impression that the empirical support for this contrast is not very strong. In the baseline specification (Eq. 1), the two team types are not statistically different and the estimated difference is small in magnitude (as captured by ω_1). While the contrast becomes statistically significant and larger in the within-pair specification, it is unclear why these two approaches should lead to substantively different conclusions if role assignment is indeed conditionally random, as assumed elsewhere in the paper.

Moreover, in the mechanisms section, once conditioning on physician C-section preferences, the analysis is split into many preference-combination cells and further decomposed by team type. In several of these preference-conditioned plots—including robustness exercises using terciles or quintiles of preferences—confidence intervals overlap substantially (especially for C-section rates),

making it difficult to rule out ML–FA = FL–MA in many cases. Even where differences appear statistically significant, the implied magnitudes seem modest in economic terms. Finally, given the large number of outcomes, specifications, and preference definitions explored, there is a risk that some statistically significant ML–FA vs FL–MA differences arise by chance in the absence of multiple-testing adjustments.

Concrete suggestions to address these points: (i) formally test how often ML–FA and FL–MA are statistically different and clarify why this should depend on the specification used (e.g. eq 1 vs within-team estimation), (ii) explicitly discuss the economic magnitude of these differences, and (iii) consider multiple-testing corrections or clearly pre-specify a small set of primary contrasts and outcomes for the gender-norms claims. If, as the evidence suggests, the two mixed-gender teams are often not statistically distinguishable and/or quite small, the paper may want to tone down claims hinging on this specific contrast and related discussion of mechanisms.

Yes I think this is sensible. Need to find a way to more seamlessly integrate the "challenging conditions" results to be less about explaining FM, but also about explaining behavior in other teams. Like for example - the "conflicting preferences" makes all teams better result (except for FM). Need to see how challenging conditions looks like with the new between and within spec though…. This may disappear.

3.2 Homogeneous teams. For mixed-gender teams, there is a well-established literature documenting backlash against female leaders and differential weighting of men's versus women's input, which provides a plausible theoretical foundation for invoking gender norms in that context. By contrast, I found the extension of a norms-based explanation to differences between all-male and all-female teams more speculative, and it is not entirely clear what specific mechanism the authors have in mind or what prior evidence supports such a channel in this setting. In particular, I found the proposed "path of least resistance" mechanism a bit loose and, as the authors themselves acknowledge, not tested empirically. If the authors want to push for this interpretation, it seems important to find sources of variation that plausibly shifts the strength or salience of norms. In general, while I found the evidence that all-female teams outperform all-male teams strong and robust, attributing this gap specifically to gender norms currently feels like a bit of a stretch.

I think it is about showing evidence that men do more intensive procedures in teams than they do alone (though this still has the solo FE issue). Also the heterogeneity results could be informative in ruling out or informing differences. Don't have to make this about gender norms? I find this the most challenging to address.

In some ways, the easiest would be "men do more aggressive interventions in teams than female physicians." While these decisions may not translate to worse outcomes in the solo sample, they do in this higher risk team sample. But don't know how to quite square this with the preferences analysis.

4. Streamline the paper

While the richness of the analysis is a strength, I also found the paper somewhat overwhelming in its current form. Results are often presented under many alternative specifications without a clear hierarchy, and the appendix is extremely long, sometimes containing material that feels essential rather than supplementary. I believe the paper would benefit from a more deliberate streamlining: selecting a single, most credible baseline specification; focusing attention on the strongest and most robust results; and bringing key supporting evidence into the main text. Doing so would greatly improve clarity and sharpen the paper's overall message.

Fair! Though not entirely sure what they want moved to the appendix vs not. The appendix can def be streamlined as the whole preferences part is a million pages.

Okay so if we use within, need to try to replicate everything with that spec. If so, then the paper would become more about MM vs MF and FM vs FF and involve major changes to the framing/narrative in ways the referees don't seem to be suggesting. Guess we will just need to see what happens!

Reviewer 2

Reviewer: 2

Comments:
Thank you very much for the opportunity to read this paper about gender diversity and team performance. The paper has an appealing empirical setting: teams of OBGYN physicians collaborating on a childbirth. The setting is appealing for multiple reasons. First, the teams consistent of only two members: a “lead” OBGYN physician and an “assisting” OBGYN physician. This means that team size is fixed, and each team has a clear hierarchy. Second, it is plausible that team formation is quasi-random, because childbirth sometimes occurs suddenly, and the lead OBGYN physician may be constrained in whom they can select to be their assistant by variation in OBGYN scheduling. Third, team performance is relatively easy to measure using maternal complications.

The authors leverage these aspects of the setting to produce a key finding: teams with two male OBGYN physicians perform the worst, while teams with two female OBGYN physicians perform the best. Mixed gender teams perform somewhere in the middle.

I think the paper has potential to provide an important contribution to the literature on gender dynamics and team performance. This is an important setting, the design and results are interesting, and the authors have carefully anticipated many issues. With these positives in mind, I have concerns. Where possible, I try to provide ideas on how the authors could address my concerns.

1. The paper seems to envision a situation where the delivery is complicated due to issues with the fetus, so another OBGYN is called. Then, later on, after delivery is over, the mother may or may not develop complications (i.e. renal failure, sepsis, etc). My main concern with the paper is that the timing of these events might not be so neat and orderly.

I am concerned that the maternal complications might **cause** the lead OBGYN to request the help of another OBGYN. In other words, there might be a kind of reverse causality or collider bias, whereby maternal complications drive formation of the team, as well as its gender mix.

This is actually ruled out by the data - the assisting obgyn needs to assist in performing a procedure, this is almost always the birth itself (we can see this in the data). They wouldn’t be on the birth record to assist with cardiac arrest or renal failure. Ie - from what we see and understand about the data, it doesn’t seem that there are many cases where the second physician would only be called in after the complication, since in most cases, they had already assisted in the delivery itself. SMM is only an outcome of delivery, it does not occur before birth (though I can double check this, get support from the clinical literature and we can look at subsets of SMM).

However, the differences in gender thresholds for calling someone based on experience can def still be a concern.

I am concerned that maternal complications might drive the gender mix of the team in part because the male doctors in the data have about 10 more years of experience than the female doctors and about 20% more births per year (Table A.1). This experience disparity, combined with the possibility that maternal complications drive team formation, cause me to worry about the following alternative story:

Higher experience (i.e. male) OBGYNs have a higher maternal complication threshold before they feel the need to ask for help from another OBGYN (e.g. if a mother is having mild complications, an OBGYN with more experience might be less likely to ask for assistance from another OBGYN). This could explain why OBGYN teams that have a man on them have higher maternal complication rates in Figure IIa. And if a high-experience (i.e. male) OBGYN faces serious maternal complications, that high-experience OBGYN will probably try hard to get assistance from another high-experience (i.e. male) OBGYN. This could explain why maternal complications are the highest for the male-male teams in Figure IIa – the causality might run the opposite way, whereby, in the team sample, the mothers with the worst complications in the are most likely to be served by male-male teams (i.e. the most experienced teams), and the mothers with the least complications are most likely to be served by female-female teams (i.e. the least experienced teams).

There are several things that the authors could do to rule out this alternative story.

a. Control for physician age and experience more effectively. ✅

Right now, in the paper controls for quintiles of physician age and cumulative births for both the lead OBGYN (Z_j in Eq 1) and the assisting OBGYN (T_k in Eq 1). It would be helpful to include an interaction of the age of the lead OBGYN and the age of the assisting OBGYN, as well as the interaction between the experience of the lead OBGYN and the assisting OBGYN. It is fine with me if these interactions use simple continuous measures of age and experience.

We can try this. When paired with R1 suggestions of including FE for shifts and day of week, this also removes the first three years of data so already taking out older male doctors. Could also just drop all doctors over a certain age.

Also age quintiles and experience are actually quite small since not huge age/experience range among Ob-Gyns.

Gently remind them how the between spec and within spec actually quite powerfully accounting for experience & decision-making to call another Ob-gyn.

b. Provide some descriptive statistics or regression analysis about “selection into a team.” ✅

It would be helpful to know whether male OBGYNs are less likely than female OBGYNs to ask for a teammate. The authors could show this in descriptive statistics and regression analysis.

It would also be helpful to know if male doctors do have a higher maternal complication threshold at which they ask for help. The authors could examine this by estimating a regression where the DV = 1 if the birth requires a team of OBGYNs and the explanatory variables include the presence or number of maternal complications and their interaction with whether or not the lead OBGYN is male.

If my story is unimportant, then we should see male and female physicians asking for a teammate at similar rates, and we should see a null interaction between presence or number of maternal complications and the gender of the lead physician.

I think we did this at one point (or something similar)? Isn't this our main endogenous team formation results? I don't quite see how we can have maternal complications on the RHS.

2. One reason I worry about my alternative explanation is that it was not clear to me why male-male teams should perform worse than male-female or female-male teams. At first, thought that maybe the issue is a lack of homophily/empathy on the part of male doctors for their female patients, but the paper asserts in Figure 1 that men and women have similar skills in single-OBGYN births, so it does not seem that men are worse OBGYNs than women. If the underlying mechanism comes from gender role dynamics and men's issues with female leadership, it was not clear to me why the male-male teams perform so poorly. ✅

3. The paper claims that "among mixed gender team, F_LEAD-M_ASSIST perform worse than M_LEAD-F_ASSIST." It was not clear to me that the data supported this conclusion. While page 18 provides p values that show that each of these teams underperform female-only teams, page 18 does not provide a formal comparison of F_LEAD-M_ASSIST to M_LEAD-F_ASSIST, and their confidence intervals seem to overlap considerably in Figure II. I know that we seem to see differences in Figure II panel c, but this analysis includes team fixed effects and might drop much of the data.

Meanwhile, the coding of the "same gender assisting variable" made it difficult for me to interpret the regression tables, which raised my concern about the previous issue. This variable pools two asymmetric comparison (female-led teams with female vs male assistants and male-led teams with male vs female assistant) while mechanically tying team composition to Lead gender.

I think it would simplify the interpretation if the authors simply use "female assisting", rather than conditioning this variable on the gender of the lead doctor. The authors could also provide p value comparisons in Figure II, like they do in later heterogeneity tests.

I think we should just do MM as leave out and report the relative effects for the other teams.

This would line up better with figures and values in the text. I think it is a bit more difficult i include p-avlue comparisons in the figure because there can be so many pairwise comparisons, but we could figure it out…

4. The authors place a lot of interpretive weight on the gender of the “lead” physician and the “assisting” physician, but, as they note, it is possible that there is a lot of noise in the assignment of physicians to these roles. It might help their conclusions if they compared their results to Monte Carlo simulations that shuffled, within cases, the identities of the lead and assisting physicians. If the results from the actual data are different from these shuffle tests, I think that it would help to strengthen the conclusions related to female versus male leadership in mixed gender teams.

R1 seemed to have a similar recommendation but for different reasons. I haven’t done this type of analysis before but seems like we will need to do it.

5. The authors cluster standard errors by lead physician. If I approach the clustering choice like the influential paper by Abadie et al. (2023) in QJE, I think I would prefer a hospital cluster, since the randomization seems to happen at the hospital level (i.e. the decision to add a teammate and the identity of the teammate will be determined by hospital protocols and staffing pool). ✅

I did this for our main results and results look pretty similar. But unclear how everything else would change.

Thank you for the opportunity to read this work, and I wish you all the best with your revisions.